

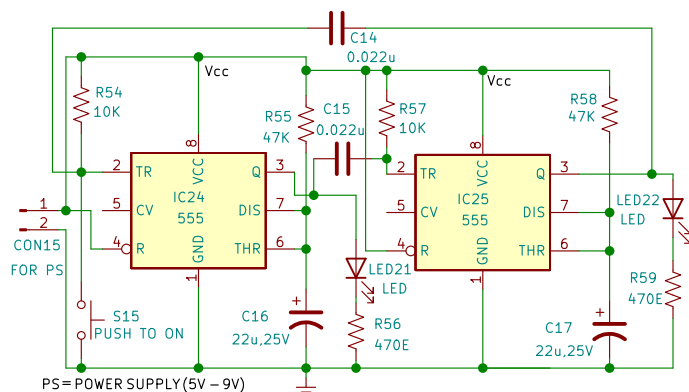


Fig. 12(a): Circuit diagram for first part of experiment 12

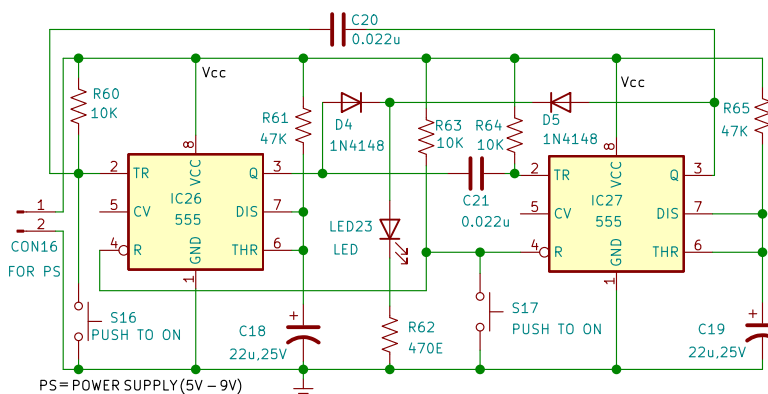


Fig. 12(b): Circuit diagram for second part of experiment 12

its one shot cycle and its output goes low and LED22 turns off.

So, although IC23 is wired as an astable multivibrator, its very first cycle remains incomplete. Thus, both the ICs together 'appear' to act as a monostable multivibrator!

Experiment 12

Circuit diagram for this experiment, shown in Fig. 12, has two parts: 12(a) and 12(b). In part 12(a) both the ICs are wired as monostable multivibrators. However, pin 3 of IC25 is connected to pin 2 of IC24 through capacitor C14 and pin 3 of IC24 is connected to pin 2 of IC25 through capacitor C15.

When switch S15 is momentar-

ily pushed, LED21 and LED22 flash alternately, as if both the ICs are wired as astable multivibrators! This is because when each monostable multivibrator's output goes low, it triggers the other monostable multivibrator and the cycle repeats.

Part 12(b) is an extension of experiment 12(a). Here also both the ICs are wired as monostable multivibrators. However, as shown in the schematic, the output of both the ICs is given to LED23 through diodes D4 and D5. Pin 4 of both the ICs are tied together and connected to Vcc through resistor R63.

When the circuit is powered and switch S16 is momentarily pushed,

PARTS LIST

Semiconductors:

- IC12-IC27 - Timer IC 555
- LED12-LED23 - 5mm LED
- D1-D5 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R24, R25, R27, R28, R30, R31, R33, R34, R36, R38, R40-R43, R45, R46, R49-R52, R54, R57, R60, R63, R64 - 10-kilo-ohm
- R26, R29, R32, R35, R39, R44, R47, R48, R53, R56, R59, R62 - 470-ohm
- R37 - 22-kilo-ohm
- R55, R58, R61, R65 - 47-kilo-ohm

Capacitors:

- C5, C9 - 10 μ F, 25V electrolytic
- C6-C8, C16-C19 - 22 μ F, 25V electrolytic
- C10 - 470 μ F, 25V electrolytic
- C11, C12, C13 - 220 μ F, 25V electrolytic

Miscellaneous:

- CON9-CON16 - 2-pin connector
- S9-S12, S14-S17 - Push-to-on switches
- S13 - SPST (ON/OFF switch)
- PS - 9V battery/0-12V variable power supply
- Breadboard

LED 23 turns on and 'appears' to glow continuously, though, in fact, it is powered alternately from output pin 3 of IC26 and IC27. Now, if switch S17 is momentarily pushed, the circuit resets and LED23 turns off.

Thus, though both IC26 and IC27 are wired as monostable multivibrators, and in Part 12(a) 'appeared' to work as astable multivibrators, now together they appear to work as a bistable multivibrator!

All these circuits can be wired on a breadboard so that the components can be reused, if required. A variable 0-12V, 1A power supply can be used for the experiments. Alternatively, a 9V battery may be used. Connectors CON9 through CON18 are used to connect power supply/battery. **EFY**



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